

Rhet O. Hailey

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Profile

I am a robotics researcher with 6+ years of experience designing, building, and controlling robots that must operate safely in close physical contact with people and constrained environments.

Education

Auburn University	Auburn, AL
Ph.D., Mechanical Engineering	Aug. 2026
<i>Evaluating Active Assessment and Learning Control for Robotic Rehabilitation</i>	
Master of Science, Mechanical Engineering	May 2023
<i>Applications of Robotic Devices for Motor and Sensory Rehabilitation</i>	
Advisor: Chad G. Rose	
Mississippi State University	Starkville, MS
Bachelor of Science, Biomedical Engineering	May 2020

Academic Appointments

Purdue University — Postdoctoral Researcher RAAD Lab	Aug. 2026 – Present
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Research Experience

Postdoctoral Research Assistant	Aug. 2026 – Present
Robust and Adaptive Design Lab at Purdue University	
Advisor: Laura Blumenschein	
Mentor graduate students and coordinate toward journal submissions and sponsor reports	
In-Situ Non-Destructive Inspection using Heterogeneous Collaborative Robots	
– Develop motion planning and control strategies for vine robots navigating confined inspection environments & led the design of soft robots with integrated sensing	
– Develop a sensor-fusion pipeline for multi-robot collaborative real-time SLAM, from simulation to physical deployment	
Graduate Research Assistant	Aug. 2020 – Aug. 2026
Wearable and Bio-Robotics Lab at Auburn University	
Advisor: Chad G. Rose	
Non-Linear Upper Extremity Exoskeleton Deep Learning Hybrid Control	
– Implemented deep neural networks in real time using Lyapunov stability for personalized adaptive control in a real-time system.	
– Developed software library for real-time network control of an exoskeleton.	
– Led implementation of real-world hybrid controllers to compare performance.	
Upper-Extremity Kinematic Assessment with Gravity Compensation for Rehabilitation	
– Collaborated with researchers at the University of Texas at Austin to conduct biomechanical analyses of subject movement wearing a gravity compensated exoskeleton.	
– Conducted kinematic analysis of the upper-extremities in post-stroke patients.	
Design of a Non-Pharmaceutical Peripheral Neuropathy Treatment Device	
– Designed and patented the Peripheral Neuropathy Treatment Device in collaboration with medical professionals to best create a comfortable, effective wearable device.	
– Analyzed magnetic resonance imaging to assess blood flow over systolic time-series.	

Journal Publications

- [J4] **Hailey, R.O.**, Ting, J., Allen, B.C., Rose, C.G. “Multi-DoF Hybrid Functional Electrical Stimulation and Upper-Extremity Exoskeleton Control,” IEEE Transactions on Medical Robotics and Bionics, pp. 1–6, 2025 (*In-review*)
- [J3] **Hailey, R.O.**, Ting, J., Allen, B.C., Rose, C.G. “L-GraD: Lyapunov-Based Gradient of a Deep Neural Network Based Upper-Extremity Exoskeleton Controller,” ASME Letters in Dynamic Systems and Control, pp. 1–6, 2025

- [J2] Baskaran, A., Hollinger, D., **Hailey, R.O.**, Zabala, M.E., Rose, C.G. “Neuromuscular State Estimation via Space-by-Time Neural Signal Decomposition,” ASME Letters in Dynamic Systems and Control, pp. 1–6, 2023
- [J1] Li, G., Zhao, Y., **Hailey, R.O.**, Zhang, N., Liang, Y., Purswell, J.L. “An Ultra-High Frequency Radio Frequency Identification System for Studying Individual Feeding and Drinking Behaviors of Group-Housed Broilers,” Animal, 2019

Refereed Conference Proceedings

- [C7] **Hailey, R.O.**, Ting, J., Mishra, K., Allen, B.C., Rose, C.G. “Assessing Hybrid Learning Controllers for Upper-Extremity Exoskeleton,” ASME Modeling Estimation and Controls Conference (MECC), 2026
- [C6] Miller, Z. R., **Hailey, R.O.**, Ting, J., Mishra, K., Allen, B.C., Rose, C.G. “Three Dimensional Analysis for Tremor Magnitude Assessment,” ASME Modeling Estimation and Controls Conference (MECC), 2026
- [C5] **Hailey, R.O.** and Rose, C.G. “Examining Postural Responses to Perturbations in 3D: A Pilot Study,” IEEE International Conference on Rehabilitation Robotics (ICORR), 2025
- [C4] **Hailey, R.O.**, Commander, C., Denney, T.S. Jr., Zabala, M.E., Burch, T., and Rose, C.G. “Tri-Modal Intervention for Diabetic Peripheral Neuropathy: Pilot Study,” The International Conference on Biomedical Robotics and Biomechatronics (BioRob), 2024
- [C3] Baskaran, A., Jamison, H., **Hailey, R.O.**, Zabala, M.E., Rose, C.G. “Design of a Haptic Paddle for Accessible Integration of Data-Driven Methods in System Dynamics Education,” IEEE International Conference on Advanced Intelligent Mechatronics (AIM), 2024
- [C2] **Hailey, R.O.**, De Oliveira, A.C., Ghonasgi, K., Rose, C.G., and Deshpande, A.D. “Impact of Gravity Compensation on Upper Extremity Movements in Harmony Exoskeleton,” IEEE International Conference on Rehabilitation Robotics (ICORR), 2022
- [C1] Li, G., Zhao, Y., **Hailey, R.O.**, Zhang, N., Liang, Y., Purswell, J.L. “Radio-Frequency Identification (RFID) System for Monitoring Specific Behaviors of Group Housed Broilers,” Proceedings of the 10th Annual ASABE Conference on International Livestock Environment Symposium (ILES X), 2018

Presentations

- [T7] “Deep Neural Network Control of Hybrid Exoskeleton Rehabilitation”, ASME DSCD Rising Stars, MECC, 10/25, Pittsburgh, PA
- [T6] “Experimental Validation of Gradient Descent Upper-extremity Exoskeleton Controller”, Rehab Week, 5/25, Chicago, IL
- [T5] “Perturbing Extremities in Robotic Tasks Uncovers Reaching Biodynamics” - IEEE BioRob, 9/24, Heidelberg, Ger,
- [T4] “Robotic Upper Extremity Task Space Perturbance: Movement Analysis” - ISMR, 6/24, Atlanta, GA
- [T3] “Real Time Fatigue Estimation for FES Controllers via Mechanomyography” - ISMR, 6/24, Atlanta, GA
- [T2] “Non-Pharmacological Peripheral Neuropathy Boot Treatment Device” - 2024 Southeast Regional Clinical & Translational Science Conference, 2/24, Pine Mountain, GA
- [T1] “Rehabilitation Training Under Gravity Compensation with Harmony Exoskeleton Improves in Movement Quality for individuals with Chronic Stroke”, Rehab Week, 7/21

Patents

- [P1] US Patent App. 19/180,660 “Method for Non-Pharmacological Treatment of Diabetic Neuropathy” Michael E. Zabala; Vitale K. Castellano; Thomas E. Burch; Jonathan C. Commander; Chad G. Rose; **Rhet O. Hailey**

Teaching Experience

Guest Lecturer:

- System Dynamics and Control (MECH 3140)
- Applied Mechatronics (MECH 6970)

Graduate Teaching Assistant:

- System Dynamics and Control (MECH 3140) (x2)
 - Organized and assessed Matlab based assignments
 - Problem working sessions
- Dynamics Laboratory (MECH 3150) (x8)
 - Organized weekly lecture series in a flipped classroom setting
 - Assisted in the design and implementation of programming based weekly projects
- Applied Mechatronics (MECH 6970) (x2)
 - Designed and constructed hardware and software for instructional device
 - Consultant towards class engagement and learning

Mentorship Experience

- Joshua Medler, BS Auburn University 2026 – Peripheral Neuropathy Assessment Automated Tool
- Zack Miller, BS Auburn University 2024 – Active Tremor Suppression Hand Wearable

Academic Honors

- Auburn University Outstanding Doctoral Student (2026)
- ASME Dynamics Systems and Controls Rising Star (2025)
- Walt and Virginia Woltosz Fellowship (2023-)
- Bagley College of Engineering Excellence Scholarship (2016-20)

Professional Society Memberships

- Member, American Society of Mechanical Engineers (ASME)
- Member, International Institute of Electrical and Electronics Engineers (IEEE)
- Member, International Consortium for Rehabilitation Robotics (ICORR)

Technical Skills

Coursework: Mechatronics, Machine Learning, Optimal Estimation, Reinforcement Learning

Hardware: Arduino, Raspberry Pis, RP2040, encoders, force/torque sensors, IMUs

Programming: C/C++/CMake/Meson, C#, LabView, Matlab, Python, ROS, Simulink, Docker

Software: KiCad, Motion capture (OptiTrak), Real-Time Linux, Solidworks, Unity, UDP/TCP

Research Administration: IRB, CITI Training, CUI